

that feed by clamping their round, jawless mouth onto the flank of a live fish, their “host.” Parasitic lampreys use their rasping mouth and tongue to penetrate the skin of the fish and ingest the fish’s blood and other tissues.

As larvae, lampreys live in freshwater streams. The larva is a suspension feeder that resembles a lancelet and spends much of its time partially buried in sediment. About 20 species of lampreys are not parasitic. These species feed only as larvae; following several years in streams, they mature sexually, reproduce, and die within a few days. In contrast, parasitic species of lampreys migrate to the sea or lakes as they mature into adults. One such parasite, the sea lamprey (*Petromyzon marinus*), has invaded the Great Lakes over the past 170 years and devastated a number of fisheries there.

The skeleton of lampreys is made of cartilage. Unlike the cartilage found in most vertebrates, lamprey cartilage contains no collagen. Instead, it is a stiff matrix of other proteins. The notochord of lampreys persists as the main axial skeleton in the adult, as it does in hagfishes. However, lampreys also have a flexible sheath around their rodlike notochord. Along the length of this sheath, pairs of cartilaginous projections related to vertebrae extend dorsally, partially enclosing the nerve cord.

Early Vertebrate Evolution

In the late 1990s, paleontologists working in China discovered a vast collection of fossils of early chordates that appear to straddle the transition to vertebrates. The fossils were formed 530 million years ago during the Cambrian explosion, when many animal groups were undergoing rapid diversification (see Concept 32.2).

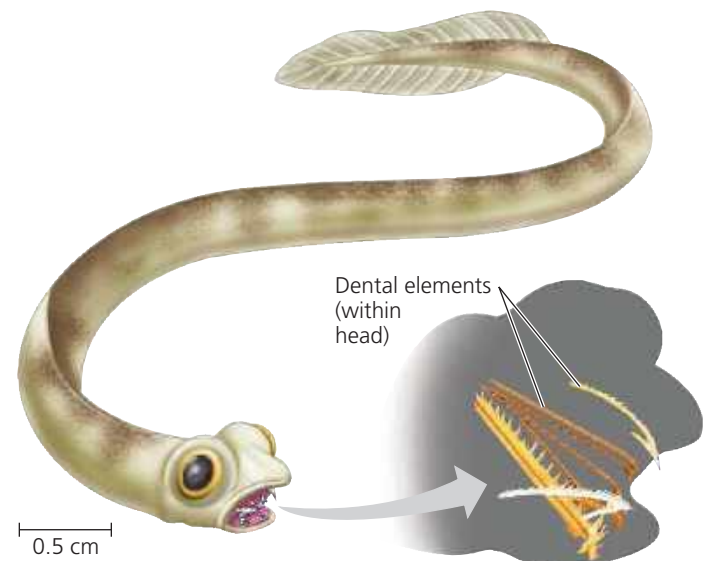
The most primitive of the fossils are the 3-cm-long *Haikouella* (Figure 34.10). In many ways, *Haikouella* resembled a lancelet. Its mouth structure indicates that, like lancelets, it probably was a suspension feeder. However, *Haikouella* also had some of the characters of vertebrates. For example, it had a well-formed brain, small eyes, and muscle segments along the body, as do the vertebrate fishes. Unlike the vertebrates, however, *Haikouella* did not have a skull or ear organs, suggesting that these characters emerged with further innovations to the chordate nervous system. (The earliest “ears” were organs for maintaining balance, a function still performed by the ears of humans and other living vertebrates.)

Early signs of a skull can be seen in *Myllokunmingia* (see Figure 34.1). About the same size as *Haikouella*, *Myllokunmingia* had ear capsules and eye capsules, parts of the skull that surround these organs. Based on these and other characters, *Myllokunmingia* is considered the first chordate to have a head. The origin of a head—consisting of a brain at the anterior end of the dorsal nerve cord, eyes and other sensory organs, and a skull—enabled chordates to coordinate more complex movement and feeding behaviors. Although it had a head, *Myllokunmingia* lacked vertebrae and hence is not classified as a vertebrate.

▼ **Figure 34.10 Fossil of an early chordate.** Discovered in 1999 in southern China, *Haikouella* had eyes and a brain but lacked a skull, a trait found in vertebrates. The organism’s color in the drawing is fanciful.



▼ **Figure 34.11 A conodont.** Conodonts were early jawless vertebrates that lived from 500 million to 200 million years ago. Unlike hagfishes and lampreys, conodonts had mineralized mouthparts, which they used for either predation or scavenging.



The earliest fossils of vertebrates date to 500 million years ago and include those of **conodonts** (Figure 34.11), a group of slender, soft-bodied vertebrates that lacked jaws and whose internal skeleton was composed of cartilage. Conodonts had large eyes, which they may have used in locating prey that